

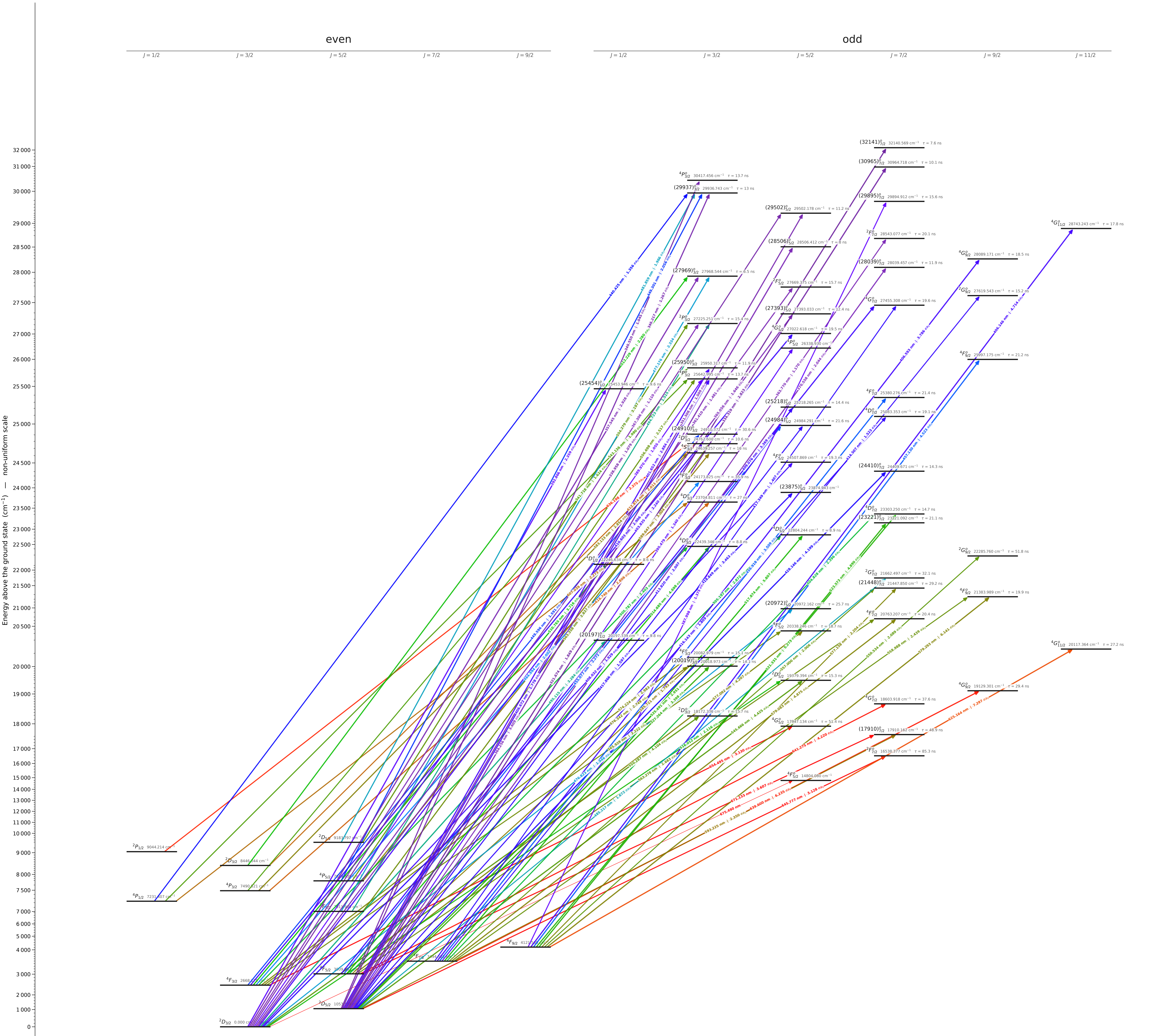
Lanthanum

La (La I). Strongest classified transitions below 2 μm (NIST): 72 levels, 90 lines.

La⁺ ionisation limit 44981.00 cm⁻¹ = 5.57693 eV (222.316 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U|er|U'|$ in ea0
Matrix elements are measured or NIST values: * = high-accuracy theory. = = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)
300 400 500 600 700 800 1000 1200 1500 2000



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U er U' $ (ea0)	A (s ⁻¹)	f/2π up. (MHz)	branch (%)	Use
² D _{3/2} - ⁴ F _{5/2}	675.4901	675.3037	443.8147		-	-	-	atomic-beam diode-laser spectroscopy of the hyperfine str...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
f/2π: natural linewidth of the upper level, 1/(2πs).

¹³⁹La hyperfine structure (I = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
² D _{3/2}	141.1959(16)	44.781(14)	F=2: -929.08 F=3: -537.48 F=4: +14.51 F=5: +752.47
² D _{5/2}	182.1706(6)	54.213(14)	F=1: -2020.38 F=2: -1671.53 F=3: -1142.44 F=4: -426.15 F=5: +486.64 F=6: +1607.55
⁴ F _{3/2}	-480.292	15.188	F=2: +3250.11 F=3: +1798.38 F=4: -127.12 F=5: -2517.74
⁴ F _{5/2}	300.5630(10)	10.873(25)	F=1: -3375.51 F=2: -2777.49 F=3: -1879.30 F=4: -679.53 F=5: +823.67 F=6: +2632.64
⁴ F _{7/2}	462.8680(10)	17.925(24)	F=0: -7280.57 F=1: -6819.53 F=2: -5897.09 F=3: -4512.51 F=4: -2664.69 F=5: -352.18 F=6: +2426.86 F=7: +5674.61
⁴ F _{9/2}	489.5340(10)	32.18(3)	F=1: -9408.78 F=2: -8434.31 F=3: -6971.65 F=4: -5019.64 F=5: -2576.76 F=6: +358.91 F=7: +3789.67 F=8: +7718.21
² F _{7/2}	146.20(20)	54(4)	F=0: -2273.99 F=1: -2133.25 F=2: -1850.67 F=3: -1424.09 F=4: -850.20 F=5: -124.66 F=6: +758.00 F=7: +1804.33
(17910) ³ F ₂	279.4(3)	86(6)	F=0: -4354.49 F=1: -4083.85 F=2: -3540.85 F=3: -2721.96 F=4: -1621.91 F=5: -233.68 F=6: +1451.49 F=7: +3444.15
⁴ G _{5/2}	539.9(8)	43(10)	F=1: -6050.84 F=2: -4983.32 F=3: -3377.45 F=4: -1227.67 F=5: +1473.36 F=6: +4734.88
² D _{5/2}	142.1(6)	-	F=2: -959.17 F=3: -532.88 F=4: +35.52 F=5: +746.02
⁴ G _{7/2}	218.8(5)	72(10)	F=0: -3407.53 F=1: -3196.08 F=2: -2771.70 F=3: -2131.46 F=4: -1270.96 F=5: -184.30 F=6: +1135.84 F=7: +2698.30
⁴ G _{9/2}	110.1(10)	85(30)	F=1: -2080.47 F=2: -1872.41 F=3: -1557.79 F=4: -1133.59 F=5: -595.73 F=6: +60.82 F=7: +842.14 F=8: +1755.33
² D _{3/2}	-58.1(6)	-	F=1: +653.62 F=2: +537.42 F=3: +363.12 F=4: +130.72 F=5: -159.78 F=6: -508.38
(20019) ³ F ₂	-41.0(3)	3(3)	F=2: +278.36 F=3: +153.21 F=4: -11.64 F=5: -214.50
⁴ F _{5/2}	87.0(5)	-2(4)	F=2: -588.32 F=3: -325.89 F=4: +22.68 F=5: +456.25
⁴ G _{7/2}	60.4(5)	100(10)	F=2: -1332.74 F=3: -1167.90 F=4: -944.48 F=5: -659.37 F=6: -308.65 F=7: +112.33 F=8: +609.03 F=9: +1187.70
(20197) ³ F ₂	132.6(20)	-	F=3: -298.35 F=4: +232.05
⁴ F _{7/2}	259.2(4)	-10(6)	F=1: -2921.36 F=2: -2400.10 F=3: -1619.29 F=4: -580.20 F=5: +715.44 F=6: +2265.50
⁴ F _{9/2}	196.2(3)	-4(3)	F=0: -3092.29 F=1: -2895.68 F=2: -2502.55 F=3: -1913.05 F=4: -1127.44 F=5: -146.03 F=6: +1030.76 F=7: +2402.45
(20972) ² F ₂	-66.4(3)	35(3)	F=1: -1765.75 F=2: -1622.95 F=3: -1412.50 F=4: -1138.90 F=5: -191.65 F=6: -572.25
⁴ F _{5/2}	93.4(3)	13(6)	F=1: -1803.91 F=2: -1615.25 F=3: -1332.85 F=4: -956.58 F=5: -487.64 F=6: +173.38 F=7: +725.55 F=8: +1467.80
(21448) ³ F ₂	121.50(20)	-35.7(13)	F=0: -1932.75 F=1: -1807.61 F=2: -1558.05 F=3: -1185.54 F=4: -692.25 F=5: -81.11 F=6: +644.25 F=7: +1479.45
² G _{7/2}	283.60(10)	58.0(20)	F=0: -4435.63 F=1: -4157.95 F=2: -3601.40 F=3: -2763.62 F=4: -1641.06 F=5: -228.98 F=6: +1478.54 F=7: +3488.60
⁴ D _{5/2}	511(3)	-	F=3: -1149.75 F=4: +894.25
² G _{9/2}	181.9(4)	56(7)	F=1: -3475.91 F=2: -3120.11 F=3: -2584.74 F=4: -1867.81 F=5: -966.64 F=6: +122.09 F=7: +1402.39 F=8: +2878.93
⁴ D _{3/2}	150(3)	-45(35)	F=2: -1033.23 F=3: -552.59 F=4: +58.27 F=5: +773.62
⁴ D _{5/2}	45(6)	-	F=1: -510.75 F=2: -419.95 F=3: -283.75 F=4: -102.15 F=5: +124.85 F=6: +397.25
(23221) ³ F ₂	-76.4(3)	59(4)	F=0: +1234.91 F=1: +1152.49 F=2: +988.85 F=3: +746.40 F=4: +428.76 F=5: +40.74 F=6: -411.64 F=7: -921.15
⁴ D _{5/2}	-28.1(5)	49(20)	F=0: +468.82 F=1: +435.73 F=2: +370.52 F=3: +275.23 F=4: +152.82 F=5: +7.33 F=6: -156.28 F=7: -331.98
⁴ D _{3/2}	105.8(10)	-	F=2: -714.15 F=3: -396.75 F=4: +26.45 F=5: +555.45
(23875) ³ F ₂	241.7(23)	-	F=1: -2719.12 F=2: -2235.72 F=3: -1510.62 F=4: -543.83 F=5: +664.67 F=6: +2114.88
⁴ F _{7/2}	-228.9(22)	30(11)	F=2: +1561.15 F=3: +853.02 F=4: -711.15 F=5: -1194.22
(24410) ³ F ₂	188.4(17)	17(15)	F=0: -2958.19 F=1: -2771.53 F=2: -2397.85 F=3: -1836.47 F=4: -1086.34 F=5: -146.07 F=6: +986.06 F=7: +2312.15
⁴ F _{5/2}	188.0(20)	15(5)	F=1: -2106.96 F=2: -1735.25 F=3: -1176.07 F=4: -427.50 F=5: +513.04 F=6: +1648.75
⁴ S ₀	-200.0(10)	3(020)	F=2: +1351.61 F=3: +749.46 F=4: -51.39 F=5: -1049.25
² D _{5/2}	418.90(20)	10(6)	F=2: -2822.22 F=3: -1572.66 F=4: +100.08 F=5: +2201.72
(24910) ³ F ₂	-300.70(20)	14(5)	F=2: +2037.22 F=3: +1125.12 F=4: -81.67 F=5: -1575.17
(24984) ³ F ₂	283.7(4)	-	F=1: -3191.62 F=2: -2624.22 F=3: -1773.12 F=4: -638.33 F=5: +780.17 F=6: +2482.38
⁴ D _{3/2}	70.8(5)	25(15)	F=0: -1101.71 F=1: -1033.46 F=2: -896.45 F=3: -689.66 F=4: -411.56 F=5: -60.12 F=6: +367.24 F=7: +873.55
(25218) ³ F ₂	101(5)	-	F=1: -1136.25 F=2: -934.25 F=3: -631.25 F=4: -227.25 F=5: +277.75 F=6: +883.75
⁴ F _{5/2}	389.0(10)	-	F=0: -6126.75 F=1: -5737.75 F=2: -4959.75 F=3: -3792.75 F=4: -2236.75 F=5: -291.75 F=6: +2042.25 F=7: +4765.25
(25454) ³ F ₂	-386.5(20)	-	F=3: +869.62 F=4: -676.38
⁴ P ₀	273.30(20)	4(4)	F=2: -1842.63 F=3: -1025.59 F=4: +66.47 F=5: +1435.83
(25950) ³ F ₂	480.3(13)	-	F=2: -3242.03 F=3: -1801.12 F=4: +120.08 F=5: +2521.57
⁴ P ₂	414.3(20)	40(20)	F=1: -2956.94 F=2: -2134.06 F=3: -5898.54 F=4: -4048.96 F=5: -2183.41 F=6: +300.49 F=7: +3205.59 F=8: +6535.23
⁴ S ₀	103.20(22)	-29(20)	F=1: -1176.54 F=2: -961.85 F=3: -642.93 F=4: -223.50 F=5: +291.46 F=6: +895.75
⁴ G _{5/2}	-147.7(16)	77(01)	F=1: +1670.73 F=2: +1370.47 F=3: +921.91 F=4: +327.23 F=5: -410.67 F=6: -1288.12
² P _{1/2}	-49.7(5)	11(6)	F=2: +341.37 F=3: -184.41 F=4: -17.53 F=5: -258.18
(27393) ³ F ₂	146.0(20)	32(14)	F=1: -1625.36 F=2: -1342.50 F=3: -914.79 F=4: -338.10 F=5: +393.04 F=6: +1285.50
⁴ G _{7/2}	358.0(16)	50(13)	F=0: -5611.71 F=1: -5258.82 F=2: -4552.00 F=3: -3489.22 F=4: -2067.43 F=5: -282.53 F=6: +1870.57 F=7: +4398.00
² G _{9/2}	186.7(20)	35(30)	F=1: -3577.93 F=2: -3209.53 F=3: -2655.89 F=4: -1915.76 F=5: -987.47 F=6: +131.07 F=7: +1442.34 F=8: +2949.28
⁴ F _{5/2}	227.0(20)	-	F=1: -2553.75 F=2: -2099.75 F=3: -1418.75 F=4: -510.75 F=5: +624.25 F=6: +1986.25
(27969) ³ F ₂	132.9(16)	-	F=2: -897.08 F=3: -498.38 F=4: +33.23 F=5: +697.73
(28039) ³ F ₂	166(6)	-	F=0: -2617.65 F=1: -2451.45 F=2: -2119.05 F=3: -1620.45 F=4: -955.65 F=5: -124.65 F=6: +872.55 F=7: +2035.95
⁴ G _{5/2}	393(5)	120(60)	F=1: -7510.25 F=2: -6741.39 F=3: -5584.54 F=4: -4035.39 F=5: -2088.25 F=6: +264.04 F=7: +3030.04 F=8: +6219.75
(28506) ³ F ₂	344(5)	-71(40)	F=1: -3911.41 F=2: -3202.53 F=3: -2146.80 F=4: -753.38 F=5: +965.59 F=6: +2994.88
² F _{7/2}	-68.5(8)	-50(30)	F=0: +1052.09 F=1: +988.69 F=2: +860.88 F=3: +666.60 F=4: +402.80 F=5: +65.41 F=6: -350.70 F=7: -851.62
⁴ G _{7/2}	374.0(020)	-	F=2: -8508.50 F=3: -7386.50 F=4: -5890.50 F=5: -4020.50 F=6: -1776.50 F=7: +941.50 F=8: +3833.50 F=9: +7199.50
(29502) ³ F ₂	49.3(5)	-11(20)	F=1: -560.52 F=2: -458.77 F=3: -307.34 F=4: -107.62 F=5: +138.48 F=6: +428.62
(29895) ³ F ₂	470.0(20)	-	F=0: -7402.50 F=1: -6932.50 F=2: -5992.50 F=3: -4582.50 F=4: -2702.50 F=5: -352.50 F=6: +2467.50 F=7: +5757.50
(29937) ³ F ₂	891(3)	-19(30)	F=2: -6027.13 F=3: -3339.36 F=4: +231.67 F=5: +4675.10
⁴ P ₀	34.5(16)	-16(30)	F=2: -241.45 F=3: -126.52 F=4: +16.05 F=5: +177.12
(30965) ³ F ₂	371.9(27)	-30(20)	F=0: -5873.50 F=1: -5498.53 F=2: -4749.23 F=3: -3626.79 F=4: -2133.07 F=5: -270.51 F=6: +1957.83 F=7: +4548.27
(32141) ³ F ₂	355.1(26)	44(20)	F=0: -5569.25 F=1: -5218.64 F=2: -4516.52 F=3: -3461.10 F=4: -2049.68 F=5: -278.67 F=6: +1856.42 F=7: +4360.98

Measured A, B constants; sources in the data file.